KAIZEN MATHS

Lesson starter

Arithmetic and Geometric Sequences and Series

Arithmetic Sequences and Series

Name

S1

For the arithmetic sequence with first term $a = -1$ and common difference $d = -4$, find u_9.

S2

In an arithmetic sequence, $u_2 = 0$ and $u_4 = 6$. Find d and a.

S3

For an arithmetic sequence with $a = -6$ and $d = 1$, find S_{10}.

S4

In an arithmetic sequence with $a = 4$ and $d = 2$, for which n is $u_n = 22$?

Show your working

KAIZEN MATHS

Lesson starter

Arithmetic and Geometric Sequences and Series

Arithmetic Sequences and Series

Name

S1

For the arithmetic sequence with first term $a = -1$ and common difference $d = -4$, find u_9.

S2

In an arithmetic sequence, $u_2 = 0$ and $u_4 = 6$. Find d and a.

S3

For an arithmetic sequence with $a = -6$ and $d = 1$, find S_{10}.

S4

In an arithmetic sequence with $a = 4$ and $d = 2$, for which n is $u_n = 22$?

ALG-B442-L1 · S

Show your working

KAIZEN MATHS

Exit ticket

Arithmetic and Geometric Sequences and Series

Arithmetic Sequences and Series

Name _____

E1

Evaluate $\sum_{k=2}^{6}(3k+3)$.

E2

For the arithmetic sequence with first term $a=-6$ and common difference $d=-1$, find u_9.

E3

In an arithmetic sequence, $u_3=-10$ and $u_6=-16$. Find d and a.

KAIZEN MATHS

Exit ticket

Arithmetic and Geometric Sequences and Series

Arithmetic Sequences and Series

Name _____

E1

Evaluate $\sum_{k=2}^{6}(3k+3)$.

E2

For the arithmetic sequence with first term $a=-6$ and common difference $d=-1$, find u_9.

E3

In an arithmetic sequence, $u_3=-10$ and $u_6=-16$. Find d and a.

ALG-B442-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Arithmetic and Geometric Sequences and Series

Arithmetic Sequences and Series

Name _____

E1

Evaluate $\sum_{k=2}^{6}(3k+3)$.

E2

For the arithmetic sequence with first term $a=-6$ and common difference $d=-1$, find u_9.

E3

In an arithmetic sequence, $u_3=-10$ and $u_6=-16$. Find d and a.

ALG-B442-L1 · E

Show your working

KAIZEN MATHS

Exit ticket

Arithmetic and Geometric Sequences and Series

Arithmetic Sequences and Series

Name _____

E1

Evaluate $\sum_{k=2}^{6}(3k+3)$.

E2

For the arithmetic sequence with first term $a=-6$ and common difference $d=-1$, find u_9.

E3

In an arithmetic sequence, $u_3=-10$ and $u_6=-16$. Find d and a.

ALG-B442-L1 · E

Show your working

Teacher answers & working

Arithmetic and Geometric Sequences and Series · Arithmetic Sequences and Series

ALG-B442-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S1

For the arithmetic sequence with first term $a = -1$ and common difference $d = -4$, find u_9.

Answer:

$u_9 = -33$

Working:

$u_n = a + (n-1)d$

$u_9 = -1 + (9-1) \cdot -4$

$\boxed{u_9 = -33}$

S2

In an arithmetic sequence, $u_2 = 0$ and $u_4 = 6$. Find d and a.

Answer:

$d = 3, \quad a = -3$

Working:

$u_n = a + (n-1)d$

$6 - 0 = (2)d$

$d = \dfrac{6-0}{2} = 3$

$a = 0 - (1) \cdot 3 = -3$

S3

For an arithmetic sequence with $a = -6$ and $d = 1$, find S_{10}.

Answer:

$S_{10} = -15$

Working:

$S_n = \dfrac{n}{2}(2a + (n-1)d)$

$S_{10} = \dfrac{10}{2}(2 \cdot -6 + (10-1) \cdot 1)$

$\boxed{S_{10} = -15}$

Teacher answers & working

Arithmetic and Geometric Sequences and Series · Arithmetic Sequences and Series

ALG-B442-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

S4

In an arithmetic sequence with $a = 4$ and $d = 2$, for which n is $u_n = 22$?

Answer:

$n = 10$

Working:

$u_n = a + (n-1)d$

$22 = 4 + (n-1) \cdot 2$

$n = \frac{18}{2} + 1 = 10$

E1

Evaluate $\sum_{k=2}^{6} (3k + 3)$.

Answer:

$= 75$

Working:

$\sum(3k+3) = 3\sum k + 3\sum 1$

$= 3 \cdot \frac{(2+6)(5)}{2} + 3 \cdot 5$

$\boxed{= 75}$

E2

For the arithmetic sequence with first term $a = -6$ and common difference $d = -1$, find u_9.

Answer:

$u_9 = -14$

Working:

$u_n = a + (n-1)d$

$u_9 = -6 + (9-1) \cdot -1$

$\boxed{u_9 = -14}$

Teacher answers & working

Arithmetic and Geometric Sequences and Series · Arithmetic Sequences and Series

ALG-B442-L1 · Student pages 1-2. Print at 100%, single-sided. Cut on dashed lines.

E3

In an arithmetic sequence, $u_3 = -10$ and $u_6 = -16$. Find d and a.

Answer:

$d = -2, \quad a = -6$

Working:

$u_n = a + (n-1)d$

$-16 - -10 = (3)d$

$d = \dfrac{-16 - -10}{3} = -2$

$a = -10 - (2) \cdot -2 = -6$